

Marked List of Content Moved to Supplementary Material and Deleted From the Main Article

*Manuscript: SPSID: A single-parameter shrinkage inverse-diffusion for denoising
gene-regulatory networks*

Prepared for the editor to address proof Queries Q1–Q2

Page reduction to meet the maximum 14-page limit for a Regular Paper

Note. All text reproduced below is quoted from the original main manuscript. Text selected for removal from the main article is highlighted in yellow. Items labelled “Moved to Supplementary Material” were removed from the main article and transferred to the standalone supplementary PDF. Items labelled “Deleted from Main Article” were removed entirely in order to reduce the page count while preserving the core methodological and validation results.

Summary of the implemented reductions

No.	Action	Original location	Disposition in revised manuscript
1	Moved to Supplementary Material	Section IV.F. Computational Performance and Scalability	Removed from the main article; supported by Supplementary Table S1.
2	Moved to Supplementary Material	Section V.D. Rank-score Comparison Across Methods, including Fig. 7 and its accompanying text	Removed from the main article and inserted into the Supplementary Materials as the new Supplementary Fig. S2 and accompanying text; all later supplementary figure numbers shifted by +1.
3	Moved to Supplementary Material	Section V.E. Validation on scRNA-seq benchmarks	Removed from the main article; full results retained in the Supplementary Materials under the renumbered supplementary figure citation.
4	Moved to Supplementary Material	Section V.F. Sensitivity to GRN incompleteness	Removed from the main article; full results retained in the Supplementary Materials under the renumbered supplementary figure citations.
5	Deleted from Main Article	Discussion paragraph beginning “A valid observation from our benchmark . . .”	Deleted to reduce repetition while retaining the main DREAM5 interpretation.
6	Deleted from Main Article	Discussion paragraph beginning “A key biological challenge is the Feed-Forward Loop (FFL) motif . . .”	Deleted because it is an additional biological interpretation paragraph rather than a core result required for the main narrative.
7	Deleted from Main Article	Discussion paragraph beginning “A key strength of SPSID is its robustness to the shrinkage parameter λ . . .”	Deleted because λ -robustness is already demonstrated in the Results section and the paragraph mainly extends future-work discussion.

Item 1

Action	Moved to Supplementary Material
Original location	Section IV.F. Computational Performance and Scalability
Disposition	Removed from the main article; retained in the Supplementary Materials via Supplementary Table S1.

Highlighted original text

IV.F. Computational Performance and Scalability

Finally, we assessed the computational cost of SPSID. The $O(n^3)$ complexity, which is standard for methods involving matrix inversion, arises from the single inversion step in Equation 10. We benchmarked the wall-clock runtime of all denoising methods on synthetic networks of increasing size n (see Supplementary Table S1 for full results). The results in Table S1 show that SPSID's absolute runtime remains practical for typical GRN sizes. On a standard desktop, SPSID takes on average 0.24 s (SD 0.02) at $n = 500$, 1.01 s (SD 0.02) at $n = 1000$, 5.33 s (SD 0.01) at $n = 2000$, 17.49 s (SD 0.17) at $n = 3000$, 48.47 s (SD 0.06) at $n = 4000$, and 148.72 s (SD 0.25) at $n = 6000$. SPSID is consistently faster than RENDOR at all sizes, and both are *orders of magnitude* faster than ND. Overall, SPSID offers competitive runtime among the higher-accuracy denoisers and remains tractable for networks up to $n = 6000$.

Explanation

This subsection reports runtime and scalability information that is supportive rather than essential to the main methodological and benchmarking narrative. The same information is fully preserved in the Supplementary Materials through Supplementary Table S1. Moving this subsection out of the main text reduces page length without affecting the presentation of the core method or the main biological validation results.

Item 2

Action	Moved to Supplementary Material
Original location	Section V.D. Rank-score Comparison Across Methods, including Fig. 7 and its accompanying text
Disposition	Removed from the main article and inserted into the Supplementary Materials as the new Supplementary Fig. S2 and accompanying text; all later supplementary figure numbers shifted by +1.

Highlighted original text

V.D. Rank-score comparison across methods

Original Fig. 7 and the full accompanying subsection text were removed from the main manuscript and transferred to the Supplementary Materials. In the revised supplementary file, former Fig. 7 is now labelled as Supplementary Fig. S2.

To assess performance consistency beyond absolute scores, we adopted a rank-based evaluation. For each of the 24 DREAM5 combinations (4 networks $\times$ 6 upstream inference methods), we ranked the M competing methods based on their AUROC and AUPR scores.

The rank $r_i \in \{1, \dots, M\}$ for a denoising method i on a given task is converted to a Rank-score (s_i) as follows:

$$s_i = 1 - \frac{r_i - 1}{M - 1}, \quad s_i \in [0, 1].$$

This score normalises the rank to the $[0, 1]$ interval, where the top-performing method receives a score of 1 and the worst receives a score of 0. This approach allows for robust aggregation of relative performance across different tasks.

Panels A and B show that SPSID attains nearly perfect mean rank-scores in both metrics and displays the narrowest inter-quartile bands, confirming that it never falls below second place for any network–inference combination. Panels C and D provide the Friedman average ranks. These rank-based assessments reinforce the earlier percentage-gain results and show that SPSID not only delivers strong average accuracy but also consistent behaviour across disparate data sets and upstream inference pipelines.

Explanation

This subsection provides an additional rank-based summary of the DREAM5 results. While informative, it is secondary to the main absolute-score analyses already presented in the main text through AUROC, AUPR, percentage-improvement analyses, and task-level visualizations. Because Fig. 7 occupies substantial space and the full subsection can be self-contained in the Supplementary Materials, relocating both the figure and its accompanying text yields a meaningful reduction in page count while preserving the analysis in full.

Item 3

Action	Moved to Supplementary Material
Original location	Section V.E. Validation on scRNA-seq benchmarks
Disposition	Removed from the main article; full results retained in the Supplementary Materials under the renumbered supplementary figure citation.

Highlighted original text

V.E. Validation on scRNA-seq benchmarks

To further validate the robustness and generalizability of SPSID, we evaluated it on seven scRNA-seq datasets curated by the BEELINE benchmark. Following the benchmark preprocessing protocol, we applied all six upstream inference methods, followed by SPSID and all five competing denoisers. The full results are presented in the Supplementary Materials. SPSID was the top-performing denoising method in the vast majority of the 42 combinations (7 datasets $\times$ 6 upstream methods), achieving the highest AUROC in 37 out of 42 tasks and the highest AUPR in 34 out of 42 tasks.

Explanation

This scRNA-seq validation is an additional robustness analysis beyond the main DREAM5 benchmark presented in the main manuscript. The complete results are preserved in the Supplementary Materials. Moving this subsection allows us to reduce length while retaining the full analysis for interested readers.

Item 4

Action	Moved to Supplementary Material
Original location	Section V.F. Sensitivity to GRN incompleteness
Disposition	Removed from the main article; full results retained in the Supplementary Materials under the renumbered supplementary figure citations.

Highlighted original text

V.F. Sensitivity to GRN incompleteness

To assess how sensitive SPSID is to incompleteness in regulatory annotations, we revisited the historical *E. coli* K-12 MG1655 regulatory networks curated in Abasy Atlas. Across snapshots, the upstream inferred networks were held fixed and identical to our DREAM5 experiments. For each backbone and snapshot pair, we applied SPSID and all competing denoisers and evaluated AUROC, AUPR, and MCC. The strong-evidence snapshot evaluation shows that SPSID's advantages persist across multiple Abasy curation releases and across diverse upstream inference backbones.

Explanation

This subsection presents an additional sensitivity analysis based on Abasy snapshot incompleteness. It is valuable as supporting evidence, but it is not necessary for establishing the main methodological contribution or the primary DREAM5 validation in the main manuscript. The full text and figures are preserved in the Supplementary Materials.

Item 5

Action	Deleted from Main Article
Original location	Discussion paragraph beginning “A valid observation from our benchmark . . .”
Disposition	Deleted to reduce repetition while retaining the main DREAM5 interpretation.

Highlighted original text

A valid observation from our benchmark is that the performance gains from SPSID are more modest when applied to already-sophisticated backbones like GENIE3, compared to the massive gains on simpler methods like Pearson correlation. This is expected, as GENIE3 itself is a powerful feature selection algorithm. However, the true utility of SPSID is demonstrated by its robustness: unlike other filters that can be unstable and catastrophically degrade strong networks, SPSID provides a stable, positive (albeit incremental) gain.

Explanation

This paragraph provides an interpretive discussion of a pattern that is already sufficiently conveyed by the results sections. Removing it helps reduce manuscript length without weakening the presentation of the method, the benchmark findings, or the main conclusion.

Item 6

Action	Deleted from Main Article
Original location	Discussion paragraph beginning “A key biological challenge is the Feed-Forward Loop (FFL) motif . . .”
Disposition	Deleted because it is an additional biological interpretation paragraph rather than a core result required for the main narrative.

Highlighted original text

A key biological challenge is the Feed-Forward Loop (FFL) motif, where a direct link ($A \rightarrow C$) co-exists with an indirect path ($A \rightarrow B \rightarrow C$). Our deconvolution framework is ideally suited for this. The observed A-C score is a mixture of the 1-step $P_{\text{dir}}(A, C)$ signal and the 2-step $P_{\text{dir}}^2(A \rightarrow B \rightarrow C)$ noise. SPSID is designed to subtract the 2-step noise component, thereby isolating the true direct link rather than eliminating it.

Explanation

This paragraph offers a useful biological interpretation, but it is not essential to the core presentation of the method or the benchmark evidence. Deleting it reduces length while leaving the main discussion intact.

Item 7

Action	Deleted from Main Article
Original location	Discussion paragraph beginning “A key strength of SPSID is its robustness to the shrinkage parameter λ . . .”
Disposition	Deleted because λ -robustness is already demonstrated in the Results section and the paragraph mainly extends future-work discussion.

Highlighted original text

A key strength of SPSID is its robustness to the shrinkage parameter λ , which allows for the use of a fixed default and removes the need for data-dependent tuning. Our analysis confirms that performance is virtually identical for any large λ . Nonetheless, a valid question is whether an adaptive parameter could yield better results in extreme network conditions.

Explanation

This paragraph discusses a possible future extension rather than a core result or necessary interpretation. It was removed to reduce page length while preserving the main discussion, the key conclusions, and the main future-direction paragraph already retained elsewhere in the manuscript.

Note

All relocated content remains available in the revised Supplementary Materials. The main manuscript has been streamlined to prioritize the core methodological development, the main simulation evidence, and the principal DREAM5 validation, while preserving additional supporting analyses in supplementary form.